Python Analysis Report:

Robinhood marketing channel performance, Jan to Jun 2025.

1. Goal

Find out which ads bring paying customers at the best cost, so the CMO can split the next six months of budget. This report shows how we explored the data, cleaned it, built the metrics, and what the numbers say.

2. The Data

- **ad_spend.csv:** 1,046 rows and 6 columns. One row for each ad on each day.
- **signups.csv:** 18,840 rows and 7 columns. One row for each new user.

ad_spend columns

- **date:** the day of the results.
- **channel:** the type of ad (Paid Search, Paid Social, Display, Affiliate).
- **campaign:** the ad name (brand_search, generic_search, prospecting_video, retargeting, display_network, finance_blogs).
- **impressions:** how many times the ad was shown.
- **clicks:** how many times people clicked it.
- **spend:** money spent that day.

signups columns

- **user_id:** a unique number for each user.
- **signup_date:** the day the user opened a free account.
- **channel:** the channel credited for the signup (includes organic and referral).
- **campaign:** the ad credited. Empty for organic and referral.
- **subscription_start_date:** the day the user started paying. Empty if they never paid.
- **plan:** monthly or yearly. Empty if they never paid.
- **revenue_first_90_days:** money earned from the user in the first 90 days.

3. Data Checks:

- **Repeated rows:** the spend file had 7 fully repeated rows. The signups file had none.

- **Channel names:** the two files wrote them differently (Paid Search and paid_search). Joining without a fix would lose rows.
- **Missing campaigns:** 4,574 signups have no campaign. We checked that all of them are organic or referral, which have no ad.
- **Free users:** 16,585 users never paid, and all of them have ₹0 revenue. So adding up revenue is safe.
- **Campaign names:** the six names are the same in both files.

4. Cleaning Steps

1. **Removed the 7 repeated spend rows.** 1,046 rows became 1,039. Repeats would count the same spend twice and make every ad look more expensive.
2. **Made a matching channel key** (lowercase, with underscores) so the two files can be joined.
3. **Made one small table for each file, with one row for each ad.** The spend file has one row for each day, and the signups file has one row for each user. Joining them directly would repeat rows.
4. **Joined the two small tables on the ad name** and checked that nothing was lost: 6 ads before and after the join, and 14,266 ad signups in both tables (18,840 total minus 4,574 organic and referral).

5. What Each Metric Means

- **Click rate:** clicks divided by views. Out of every 100 views, how many were clicked.
- **Cost of one click:** money spent divided by clicks.
- **Cost of one signup:** money spent divided by signups.
- **Cost for each paying customer:** money spent divided by the number of people who paid. This is the most important one, because the company earns money only from paying customers.
- **Pay rate:** paying customers divided by signups. Out of every 100 signups, how many pay.
- **Money back per ₹1 (ROAS):** money earned divided by money spent. Below 1.00 means the ad lost money in the first 90 days.

Every ratio is worked out from the totals, not by averaging daily ratios. A very small day would otherwise count as much as a very big day.

6. The Metrics Table

This table is saved as campaign_metrics.csv.

Money (first 90 days)

Ad	Money spent	Money earned	Result
brand_search	₹9,84,550	₹18,69,137	+₹8,84,587
retargeting	₹12,54,671	₹16,71,494	+₹4,16,823
generic_search	₹34,67,901	₹16,26,299	−₹18,41,602
finance_blogs	₹14,86,954	₹6,06,375	−₹8,80,579
prospecting_video	₹32,89,884	₹8,41,481	−₹24,48,403
display_network	₹15,93,460	₹51,828	−₹15,41,632

Signups, payers and cost

Ad	Signups	Paying	Pay rate	Cost of one signup	Cost for each paying customer	Money back per ₹1
brand_search	2,177	465	21.4%	₹452	₹2,117	₹1.90
retargeting	2,465	418	17.0%	₹509	₹3,002	₹1.33
generic_search	2,008	432	21.5%	₹1,727	₹8,028	₹0.47
finance_blogs	864	162	18.8%	₹1,721	₹9,179	₹0.41
prospecting_video	6,484	213	3.3%	₹507	₹15,445	₹0.26
display_network	268	8	3.0%	₹5,946	₹1,99,182	₹0.03

Clicks

Ad	Views	Clicks	Click rate	Cost of one click
brand_search	2,69,367	12,173	4.52%	₹81
retargeting	10,92,376	17,373	1.59%	₹72
generic_search	9,02,920	22,270	2.47%	₹156
finance_blogs	7,10,919	7,794	1.10%	₹191

Ad	Views	Clicks	Click rate	Cost of one click
prospecting_video	57,20,376	68,774	1.20%	₹48
display_network	66,60,323	7,907	0.12%	₹202

7. Paid Channels vs Free Channels

Out of every 100 signups, how many start paying:

- paid_search: 21
- referral: 20
- affiliate: 19
- organic: 10
- paid_social: 7
- display: 3

The paid_social number mixes two very different ads: retargeting (about 17 out of 100 pay) and prospecting_video (about 3 out of 100 pay).

8. Six-Month Trend

- **display_network stopped.** It spent about ₹3.4 to ₹3.7 lakh a month until April, about ₹1.6 lakh in May, and nothing in June. The data does not say why.
- **The other five ads spent about the same every month.**
- **The cost for each paying customer was steady:** brand_search ₹1,900 to ₹2,300, retargeting ₹2,800 to ₹3,300, generic_search ₹7,300 to ₹9,500, finance_blogs ₹7,800 to ₹11,000. prospecting_video was ₹21,800 in January, then ₹14,200 to ₹15,100.
- **display_network had no paying customers in March and May.**

9. Findings

1. **brand_search is the best ad.** It is the cheapest way to get a paying customer (₹2,117) and earns ₹1.90 back for every ₹1.
2. **retargeting is second** (₹3,002, and ₹1.33 back). These are the only two ads that earned back more than they cost.

3. **Cheap signups are not cheap paying customers.** prospecting_video has the 2nd cheapest signup (₹507), but the 5th cheapest paying customer (₹15,445). Only about 3 out of 100 of its signups pay.
4. **generic_search moves up** from 5th for the cost of a signup to 3rd for the cost of a paying customer. Its signups pay as often as brand_search signups.
5. **display_network is the worst:** ₹1,99,182 for each paying customer, and almost nobody clicks it.
6. **Judge ads by the cost of a paying customer,** not by the cost of a signup.

10. Limits and Notes

- Only the first 90 days of revenue are counted. Some customers keep paying longer, so slower ads may look worse than they are.
- Each signup is linked to only one ad.
- June signups have had little time to start paying.
- These results show what happened, not why.
- The SQL and Power BI spend totals (₹1.20 crore) are a little under 1% below this report (about ₹1.21 crore). The cause was not found, and no ranking changes.

11. Files from This Analysis

- Python notebook
- campaign_metrics.csv (the three tables in section 6)